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Developing device and image forming apparatus having operation mode for cleaning

Abstract

A developing device includes a developer holder that supplies developer to an image carrier on which an electrostatic latent image is formed. The developing device further includes a removing member that removes the developer that has not been supplied to the image carrier and remains on the developer holder and a transport unit that includes a helical rotating body, the transport unit rotating the rotating body to transport the developer and supply the developer to the developer holder. The developing device further includes an accommodating unit that accommodates the developer holder, the removing member, and the transport unit; and a drive mechanism that operates the developer holder and the transport unit together in a first operation mode for image formation and operates the developer holder and the transport unit individually in a second operation mode for cleaning an inside of the accommodating unit by removing the developer.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
(1) This application is based on and claims priority under 35 USC 119 from Japanese Patent Application No. 2023-158338 filed Sep. 22, 2023.
BACKGROUND
(i) Technical Field
(2) The present disclosure relates to a developing device and an image forming apparatus.
(ii) Related Art
(3) An electrophotographic image forming apparatus typically uses developer containing toner and carrier. This type of developer gradually deteriorates with use, and the deterioration of the developer leads to a reduction in the image quality. Therefore, the developer in the developing device needs to be replaced regularly.
(4) Japanese Unexamined Patent Application Publication No. 6-230668 describes a method for discharging developer from a container. This method includes opening a developer discharge port; repeating reverse rotation of transport units (transport screws) for discharging the developer

accumulated at the bottom of the container and forward rotation of a developer carrier for scraping off the developer on the developer carrier to the bottom of the container with a separating member; and then performing reverse rotation of the transport units to discharge the developer that has been scraped off.

SUMMARY

(5) When the transport units are operated together with the developer holder in the process of removing the developer from the developer holder, the developer is supplied from the transport units to the developer holder. The developer supplied from the transport units adheres to the developer holder, and therefore the developer discharge efficiency is reduced.

(6) Aspects of non-limiting embodiments of the present disclosure relate to a structure for increasing the developer discharge efficiency compared to when the transport units operate together with the developer holder in the process of discharging the developer.

(7) Aspects of certain non-limiting embodiments of the present disclosure overcome the above disadvantages and/or other disadvantages not described above. However, aspects of the non-limiting embodiments are not required to overcome the disadvantages described above, and aspects of the non-limiting embodiments of the present disclosure may not overcome any of the disadvantages described above.

(8) According to an aspect of the present disclosure, there is provided a developing device including developer holder that supplies developer to an image carrier on which an electrostatic latent image is formed; a removing member that removes the developer that has not been supplied to the image carrier and remains on the developer holder; a transport unit that includes a helical rotating body, the transport unit rotating the rotating body to transport the developer and supply the developer to the developer holder; an accommodating unit that accommodates the developer holder, the removing member, and the transport unit; and a drive mechanism that operates the developer holder and the transport unit together in a first operation mode for image formation and operates the developer holder and the transport unit individually in a second operation mode for cleaning an inside of the accommodating unit by removing the developer.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) An exemplary embodiment of the present disclosure will be described in detail based on the following figures, wherein:

(2) FIG. **1** illustrates an image forming apparatus to which the exemplary embodiment is applied;

(3) FIG. **2** illustrates the structure of a relevant part of a developing unit;

(4) FIG. **3** illustrates the overall structure of the developing unit;

(5) FIG. **4** illustrates the operation of first to third transport units in a second operation mode;

(6) FIG. **5** is a flowchart of an operation of the developing unit in the second operation mode;

(7) FIG. **6** illustrates an example of the structure of a drive mechanism that distributes a driving force when a solenoid switch is OFF; and

(8) FIG. **7** illustrates the structure of the drive mechanism that distributes the driving force when the solenoid switch is ON.

DETAILED DESCRIPTION

(9) An exemplary embodiment of the present disclosure will now be described in detail with reference to the accompanying drawings.

Structure of Image Forming Apparatus

(10) FIG. **1** illustrates an image forming apparatus **1** to which the present exemplary embodiment is applied. The image forming apparatus **1** according to the present exemplary embodiment includes a paper feed unit **1**A, a print unit **1**B, and a paper output unit **1**C. The paper feed unit **1**A includes

first to fourth paper storage members **11** to **14** that store paper sheets P serving as examples of recording media. The paper feed unit **1**A includes feed rollers **15** to **18** provided for the first to fourth paper storage members **11** to **14**, respectively, to deliver the paper sheets P stored in the respective paper storage members to a transport path connected to the print unit **1**B.
(11) The print unit **1**B includes an image forming section **20** in which an image is formed on each paper sheet P. The print unit **1**B also includes a controller **21** that controls components of the image forming apparatus **1**. In the present exemplary embodiment, the controller **21** controls the operation of each developing unit **33** of the image forming section **20** described below in a first operation mode, which is an operation mode for image formation, and a second operation mode, which is an operation mode for cleaning the developing unit **33**. The operation performed in each operation mode will be described in detail below. The print unit **1**B also includes an image processor **22**. The image processor **22** performs an image process on image data transmitted from an image reading device **4** or a personal computer (PC) **5**. The print unit **1**B also includes a user interface (UI) **23** composed of, for example, a touch panel for presenting information to a user and receiving information from the user.
(12) The image forming section **20**, which is an example of image forming means, includes six image forming units **30**T, **30**P, **30**Y, **30**M, **30**C, and **30**K (hereinafter sometimes referred to simply as “image forming units **30**”) arranged in parallel with uniform spacing therebetween. Each image forming unit **30** includes a photoconductor drum **31** on which an electrostatic latent image is formed while the photoconductor drum **31** rotates in the direction of arrow A; a charging roller **32** that charges a surface of the photoconductor drum **31**; the developing unit **33** that develops the electrostatic latent image formed on the photoconductor drum **31**; and a drum cleaner **34** the removes toner and the like from the surface of the photoconductor drum **31**.
(13) The image forming section **20** also includes an exposure device **26** that exposes the photoconductor drum **31** of each image forming unit **30** to laser light. The light to which the photoconductor drum **31** is exposed by the exposure device **26** is not limited to laser light. For example, a light source, such as a light emitting diode (LED), may be provided for each image forming unit **30**, and the photoconductor drum **31** may be exposed to light emitted from the light source.
(14) The image forming units **30** have the same structure except for the toner contained in the developing units **33**. The image forming units **30**Y, **30**M, **30**C, and **30**K respectively form yellow (Y), magenta (M), cyan (C), and black (K) toner images. The image forming units **30**T and **30**P form toner images using, for example, toner of a corporate color, foaming toner for printing Braille characters, toner of a fluorescent color, or toner used to improve glossiness. In other words, the image forming units **30**T and **30**P form toner images using toners of special colors.
(15) The image forming section **20** also includes an intermediate transfer belt **41** to which the toner images of respective colors formed on the photoconductor drums **31** of the image forming units **30** are transferred. The image forming section **20** also includes first transfer rollers **42** that transfer the toner images of the respective colors formed by the image forming units **30** onto the intermediate transfer belt **41** in first transfer regions T**1**. The image forming section **20** also includes a second transfer roller **40** that simultaneously transfers the toner images transferred to the intermediate transfer belt **41** to the paper sheet P in a second transfer region T**2**. The image forming section **20** also includes a belt cleaner **45** that removes toners and the like from a surface of the intermediate transfer belt **41**, and a fixing device **80** that fixes the images transferred to the paper sheet P in the second transfer process to the paper sheet P.
(16) The image forming section **20** performs an image forming operation based on a control signal transmitted from the controller **21**. More specifically, first, the image forming section **20** causes the image processor **22** to perform the image process on the image data received from the image reading device **4** or the PC **5** and supply the resulting image data to the exposure device **26**. Then, in, for example, the magenta (M) image forming unit **30**M, the charging roller **32** charges the

surface of the photoconductor drum **31**, and the exposure device **26** irradiates the photoconductor drum **31** with laser light modulated based on the image data obtained from the image processor **22**.
(17) Thus, an electrostatic latent image is formed on the photoconductor drum **31**. The formed electrostatic latent image is developed by the developing unit **33**, so that a magenta toner image is formed on the photoconductor drum **31**. Similarly, the image forming units **30**Y, **30**C, and **30**K respectively form yellow, cyan, and black toner images, and the image forming units **30**T and **30**P form toner images of special colors.
(18) The toner images of the respective colors formed by the image forming units **30** are successively electrostatically transferred to the intermediate transfer belt **41** rotating in the direction of arrow C in FIG. **1** by the first transfer rollers **42**, so that the toner images are superposed on the intermediate transfer belt **41**. The superposed toner images on the intermediate transfer belt **41** are transported toward the second transfer region T**2** including the second transfer roller **40** and a backup roller **49** as the intermediate transfer belt **41** moves.
(19) The paper sheet P is fed from, for example, the first paper storage member **11** by the feed roller **15**, and then is transported to the position of the registration roller **74** along the transport path. The registration roller **74** supplies the paper sheet P to the second transfer region T**2** at the time when the superposed toner images are transported to the second transfer region T**2**. The superposed toner images are simultaneously electrostatically transferred to the paper sheet P by a transferring electric field formed between the second transfer roller **40** and the backup roller **49** in the second transfer region T**2**.
(20) After that, the paper sheet P to which the superposed toner images have been electrostatically transferred is transported to the fixing device **80**. The fixing device **80** performs a fixing process in which the paper sheet P having the unfixed toner images formed thereon is heated and pressed so that the toner images are fixed to the paper sheet P. The paper sheet P that has undergone the fixing process passes through a paper straightening section **81** provided in the paper output unit **1**C, and is transported to a paper stacking portion (not illustrated).
Structure of Developing Unit **33**
(21) FIG. **2** illustrates the structure of the developing unit **33**. FIG. **2** is a perspective view of the developing unit **33** viewed in the direction from the front (near side of FIG. **1**) to the rear (far side of FIG. **1**) of the print unit **1**B included in the image forming apparatus **1** illustrated in FIG. **1**, and illustrates a sectional view taken at a location near the front end.
(22) The developing unit **33** includes an accommodating unit **331** disposed adjacent to the photoconductor drum **31** and composed of a housing extending in the front-to-rear direction of the print unit **1**B. The accommodating unit **331** has an opening **331***a* at a position facing the photoconductor drum **31**. The accommodating unit **331** accommodates a developing roller **332**, a pickup roller **333**, a first transport unit **334**, a second transport unit **335**, a third transport unit **336**, a regulating member **337**, and a removing member **338** together with developer. The members accommodated in the accommodating unit **331** are elongated members having similar lengths, and are disposed substantially parallel to each other. The developer contained in the accommodating unit **331** is a mixture of toner used to form an image and carrier composed of magnetic particles for carrying the toner.
(23) The developing roller **332** includes a solid cylindrical shaft member **332***a* and a hollow cylindrical sleeve **332***b* that covers the shaft member **332***a*. The developing roller **332** is disposed such that a side surface of the sleeve **332***b* is partially exposed at the opening **331***a* in the accommodating unit **331** and that the exposed surface faces the photoconductor drum **31**. The shaft member **332***a* exerts a magnetic force. The sleeve **332***b* rotates around the shaft member **332***a* in the direction shown by the arrow. The developing roller **332** causes the developer to adhere to the surface of the sleeve **332***b* in response to the magnetic force of the shaft member **332***a*, and rotates the sleeve **332***b* to convey the developer to the opening **331***a*, so that the toner adheres to the charged photoconductor drum **31** to develop an electrostatic latent image. The developing roller

332 is an example of a developer holder.
(24) The pickup roller **333** is disposed adjacent and parallel to the developing roller **332**. The pickup roller **333** includes a solid cylindrical shaft member **333***a* and a hollow cylindrical sleeve **333***b* that covers the shaft member **333***a*. The shaft member **333***a* exerts a magnetic force. The sleeve **333***b* rotates around the shaft member **333***a*. The pickup roller **333** causes the developer to adhere to the surface of the sleeve **333***b* in response to the magnetic force of the shaft member **333***a*, and rotates the sleeve **333***b* to convey and transfer the developer to the developing roller **332**. The pickup roller **333** is an example of a developer relay unit.
(25) Each of the first transport unit **334**, the second transport unit **335**, and the third transport unit **336** is an auger having a helical blade around a rotating shaft. The first transport unit **334**, the second transport unit **335**, and the third transport unit **336** are disposed in the accommodating unit **331** such that rotating shafts thereof are parallel to the developing roller **332** and the pickup roller **333**. The first transport unit **334**, the second transport unit **335**, and the third transport unit **336** are rotated around the rotating shafts thereof so that the developer is stirred and transported in the directions along the rotating shafts of the transport units **334** to **336**. The developer is transported along the first transport unit **334**, the second transport unit **335**, and the third transport unit **336** in that order as the first transport unit **334**, the second transport unit **335**, and the third transport unit **336** rotate, and is supplied from the third transport unit **336** to the pickup roller **333**. The direction in which the developer is transported to be supplied to the pickup roller **333** in the image forming operation is referred to as a “forward direction”.
(26) The regulating member **337** is disposed downstream of the position at which the sleeve **332***b* faces the pickup roller **333** and upstream of the position at which the developing roller **332** faces the photoconductor drum **31** along the side surface of the developing roller **332**. Here, the terms “downstream” and “upstream” respectively refer to downstream and upstream in the direction in which the sleeve **332***b* of the developing roller **332** rotates. The regulating member **337** levels off the developer supplied by the pickup roller **333** and held on the surface of the sleeve **332***b* of the developing roller **332** so that the height of the developer is constant. The developer removed by the regulating member **337** is collected in the accommodating unit **331** and returned to the transport path composed of the first transport unit **334**, the second transport unit **335**, and the third transport unit **336**.
(27) The removing member **338** is disposed downstream of the position at which the sleeve **332***b* faces the photoconductor drum **31** along the side surface of the developing roller **332**. The term “downstream” refers to downstream in the direction in which the sleeve **332***b* of the developing roller **332** rotates. The removing member **338** scrapes off the developer remaining on the surface of the sleeve **332***b* after the developer is supplied from the developing roller **332** to the photoconductor drum **31**. The developer removed from the developing roller **332** by the removing member **338** is collected in the accommodating unit **331** and returned to the transport path composed of the first transport unit **334**, the second transport unit **335**, and the third transport unit **336**.
Discharge and Introduction of Developer
(28) FIG. **3** illustrates the overall structure of the developing unit **33**. The developing unit **33** extends in the front-to-rear direction of the print unit **1**B. The developing unit **33** includes a frame **330** that supports the accommodating unit **331** and the developing roller **332**, the pickup roller **333**, the first to third transport units **334** to **336**, the regulating member **337**, and the removing member **338** accommodated in the accommodating unit **331**.
(29) The accommodating unit **331** has a discharge port **331***b* and a filling port **331***c* that are capable of being opened and closed at the front end thereof. The discharge port **331***b* is opened when the developer is to be replaced due to deterioration of the carrier contained in the developer, and the developer is discharged from the inside of the accommodating unit **331** through the discharge port **331***b*. The filling port **331***c* is opened when the developing unit **33** is to be filled with the developer

or toner, and the developer or toner is introduced into the accommodating unit **331** through the filling port **331***c*. The operation of opening and closing the discharge port **331***b* and the filling port **331***c* is controlled by for example, the controller **21** (see FIG. **1**).
(30) When the image forming section **20** (see FIG. **1**) of the print unit **1**B forms an image, the toner is supplied form the developing roller **332** to the photoconductor drum **31** in the developing unit **33**, so that a toner image is formed. Thus, the toner is consumed and the toner concentration in the developer contained in the accommodating unit **331** of the developing unit **33** is reduced. Therefore, for example, when the toner concentration in the developer contained in the accommodating unit **331** becomes less than or equal to a predetermined concentration, the filling port **331***c* is opened and toner is supplied through the filling port **331***c* from a hopper (not illustrated).
(31) When the image forming section **20** forms an image, the carrier contained in the developer in the accommodating unit **331** is deteriorated by, for example, wear. Therefore, the developer in the accommodating unit **331** is replaced when, for example, the number of times the print unit **1**B has been operated reaches or exceeds a predetermined number. In this case, first, the discharge port **331***b* is opened and the developer is discharged from the inside of the accommodating unit **331**. Then, the discharge port **331***b* is closed and the filling port **331***c* is opened to allow new developer to be introduced into the accommodating unit **331**.
(32) When the developer is to be discharged from the inside of the accommodating unit **331**, the developing unit **33** performs an operation different from that for image formation to clean the inside of the accommodating unit **331**, and thereby discharges a large amount of developer. More specifically, in the second operation mode, which is an operation mode for cleaning, the developing unit **33** operates the developing roller **332**, the pickup roller **333**, and the first to third transport units **334** to **336** individually. The developing unit **33** includes a drive mechanism capable of operating these components individually.
Drive Mechanism for Developing Unit **33**
(33) As illustrated in FIG. **3**, the drive mechanism of the developing unit **33** includes a first drive unit **51**, a second drive unit **52**, and a third drive unit **53**. The first drive unit **51** includes a motor for rotating the developing roller **332**, and is disposed at a position corresponding to the position of the developing roller **332** at the rear of the developing unit **33** in the example illustrated in FIG. **3**. The second drive unit **52** includes a motor for rotating the pickup roller **333**, and is disposed at a position corresponding to the position of the pickup roller **333** at the front of the developing unit **33** in the example illustrated in FIG. **3**. The third drive unit **53** includes a motor for rotating the first to third transport units **334** to **336**, and is disposed at a position corresponding to the position of the first to third transport units **334** to **336** at the rear of the developing unit **33** in the example illustrated in FIG. **3**. The third drive unit **53** is assumed to simultaneously drive the first to third transport units **334** to **336** using one motor, but may have a structure including plural motors or a structure including gears and clutches for changing driven objects so that the first to third transport units **334** to **336** may be driven individually.
Operation of Developing Unit **33**
(34) The developing unit **33** operates in either the first operation mode, which is an operation mode for normal image formation, or the second operation mode, which is an operation mode for cleaning the inside of the accommodating unit **331**, for example, for replacement of the developer. The operation mode is switched between the first operation mode and the second operation mode by, for example, the controller **21** (see FIG. **1**).
(35) In the first operation mode, the developing unit **33** causes the first to third drive units **51** to **53** to operate the developing roller **332**, the pickup roller **333**, and the first to third transport units **334** to **336** simultaneously. Accordingly, the developer contained in the accommodating unit **331** of the developing unit **33** is transported by the first to third transport units **334** to **336** to the pickup roller **333** and then supplied to the developing roller **332**. The developing roller **332** forms a toner image

on the photoconductor drum **31**. The developer remaining on the developing roller **332** after the image is developed on the photoconductor drum **31** is removed by the removing member **338** and returned to the transport path composed of the first to third transport units **334** to **336** in the accommodating unit **331**.

(36) In the second operation mode, the developing unit **33** individually causes the first drive unit **51** to drive the developing roller **332**, the second drive unit **52** to drive the pickup roller **333**, and the third drive unit **53** to drive the first to third transport units **334** to **336**. In the cleaning operation, the developing roller **332** is operated so that the developer on the surface of the developing roller **332** is scraped off by the removing member **338**. In this operation, when the pickup roller **333** and the first to third transport units **334** to **336** are also operated, the developing roller **332** from which the developer has been scraped off receives new developer supplied thereto, and the new developer adheres to the surface of the developing roller **332**. Therefore, in the second operation mode, the developing roller **332**, the pickup roller **333**, and the first to third transport units **334** to **336** are operated individually so that the developer is not supplied to the member from which the developer is removed.

(37) The operation in the second operation mode may be performed in accordance with, for example, the following procedure. First, the pickup roller **333** is operated while the first to third transport units **334** to **336** are stopped so that the developer on the pickup roller **333** is transferred to the developing roller **332**, and the developing roller **332** is operated to remove the developer on the developing roller **332**. After that, the first to third transport units **334** to **336** are operated while the pickup roller **333** and the developing roller **332** are stopped, so that the developer is discharged through the discharge port **331***b*. The operation of the first to third transport units **334** to **336** in the second operation mode for discharging the developer differs from the operation of the first to third transport units **334** to **336** in the first operation mode for image formation.

(38) FIG. **4** illustrates the operation of the first to third transport units **334** to **336** in the second operation mode. FIG. **4** is the bottom view of the first to third transport units **334** to **336** of the developing unit **33** seen through the accommodating unit **331**, and only the second transport unit **335** and the third transport unit **336** at the lowermost location are illustrated. Although the first transport unit **334**, which is positioned above the second transport unit **335** and the third transport unit **336**, is not illustrated in FIG. **2**, the first transport unit **334** operates in association with the second transport unit **335** and the third transport unit **336**. Other members accommodated in the accommodating unit **331** are also not illustrated in FIG. **4**. In FIG. **4**, the discharge port **331***b* provided in the accommodating unit **331** (not illustrated) is shown by the dashed lines.

(39) In the second operation mode, the first to third transport units **334** to **336** repeat an operation of transporting the developer in the forward direction as in the first operation mode and an operation of transporting the developer in a direction opposite to the forward direction (hereinafter referred to as a “reverse direction”) multiple times. When the developer accumulates on the bottom of the accommodating unit **331**, the developer agglomerates and forms a layer (residual layer). In the second operation mode, the first to third transport units **334** to **336** repeat the operation in the forward direction and the operation in the reverse direction to disintegrate the residual layer and transport the disintegrated developer.

(40) FIG. **5** is a flowchart showing the operation of the developing unit **33** in a second operation mode. When the operation in the second operation mode is started to replace the developer in the developing unit **33** and clean the inside of the accommodating unit **331** (S**101**), the discharge port **331***b* of the accommodating unit **331** is opened (S**102**), and the developing roller **332**, the pickup roller **333**, and the first to third transport units **334** to **336** are driven by the first to third drive units **51** to **53** in the second operation mode.

(41) In the operation of the first to third drive units **51** to **53** in the second operation mode, first, the second drive unit **52** drives the pickup roller **333** in the forward direction (S**103**). Accordingly, the developer on the pickup roller **333** is transfer to the developing roller **332**. At this time, only the

pickup roller **333** is operated, and the developing roller **332** and the first to third transport units **334** to **336** are stopped. Therefore, the developer is not supplied from the first to third transport units **334** to **336** to the pickup roller **333**.

(42) Next, the first drive unit **51** drives the developing roller **332** in the forward direction (S**104**). Accordingly, the developer on the developing roller **332** is scraped off by the regulating member **337** and the removing member **338**. At this time, only the developing roller **332** is operated, and the pickup roller **333** and the first to third transport units **334** to **336** are stopped. Therefore, the developer is not supplied from the pickup roller **333** to the developing roller **332**.

(43) Next, the third drive unit **53** repeatedly drives the first to third transport units **334** to **336** in the forward and reverse directions multiple times. More specifically, first, the third drive unit **53** drives the first to third transport units **334** to **336** in the forward direction for a certain period of time (S**105**), and then drives the first to third transport units **334** to **336** in the reverse direction for a certain period of time (S**106**). The third drive unit **53** repeats this operation a predetermined number of times (S**107**). The third drive unit **53** repeatedly drives the first to third transport units **334** to **336** in the forward direction (S**105**) and in the reverse direction (S**106**) while the predetermined number of times is not reached (NO in S**107**), and stops driving the first to third transport units **334** to **336** when the predetermined number of times is reached (YES in S**107**). Although an execution time of the operation for each direction is not particularly limited, since the operation is performed to discharge the developer in the accommodating unit **331**, the execution time may be set to a time in which the developer is transported from one end to the other end of one transport unit (for example, the third transport unit **336**) and may be, for example, about 20 seconds. The number of times the driving in the forward direction and the driving in the reverse direction are repeated is not particularly limited and may be, for example, three or four.

(44) After the operation in which the third drive unit **53** drives the first to third transport units **334** to **336** is ended, the discharge port **331***b* is closed (S**108**). Then, the filling port **331***c* is opened, and a container (for example, a hopper) containing the developer is set to the filling port **331***c* (S**109**). After that, the first to third drive units **51** to **53** drive the developing roller **332**, the pickup roller **333**, and the first to third transport units **334** to **336** in the forward direction (S**110**).

(45) Thus, the developer is introduced into the accommodating unit **331** through the filling port **331***c*, and replacement of the developer is completed. The operation described above with reference to FIG. **5** is merely an example of the operation of the developing unit **33** in the second operation mode, and the operation in the second operation mode is not limited to the procedure illustrated in FIG. **5** as long as the developer may be removed from the developing roller **332** and the pickup roller **333** while the supply of the developer from the first to third transport units **334** to **336** is stopped. For example, in the above-described example, the pickup roller **333** is operated, and then the developing roller **332** is operated (S**103** and S**104**). However, when the first to third transport units **334** to **336** are stopped, the developing roller **332** and the pickup roller **333** may be operated simultaneously. When the first to third transport units **334** to **336** are driven in the forward and reverse directions (S**105** to **107**), the developing roller **332** may also be operated because the developer is not supplied to the developing roller **332** as long as the pickup roller **333** is stopped.

Modification

(46) In the above-described example, the developing unit **33** includes the developing roller **332**, the pickup roller **333**, and the first to third transport units **334** to **336**. The developing unit **33** may include no pickup roller **333** depending on the type thereof. In such a structure, the developer transported by the first to third transport units **334** to **336** is directly supplied from the third transport unit **336** to the developing roller **332**. When the developing unit **33** having such a structure operates in the second operation mode (see FIG. **5**), S**103** is omitted after the discharge port **331***b* is opened in S**102**, and the developing roller **332** is driven first (S**104**). Then, the first to third transport units **334** to **336** are driven (S**105** to S**107**), the discharge port is closed (S**108**), and the filling port **331***c* is opened to introduce the developer (S**109** and S**110**).

(47) In the developing unit **33** having the above-described structure, the first to third drive units **51** to **53** of the drive mechanism include the respective motors to individually drive the developing roller **332**, the pickup roller **333**, and the first to third transport units **334** to **336**. In contrast, the drive mechanism may include one drive source (motor) and a mechanism including gears for transmitting the driving force, and be configured to distribute the driving force between the developing roller **332**, the pickup roller **333**, and the first to third transport units **334** to **336**.
(48) FIGS. **6** and **7** illustrate an example of the structure of a drive mechanism that distributes the driving force. In the example illustrated in FIGS. **6** and **7**, the developing unit **33** includes no pickup roller **333**, and includes only the developing roller **332** and the first to third transport units **334** to **336**. In this case, in the second operation mode, the developing roller **332** and the first to third transport units **334** to **336** operate individually.
(49) A drive mechanism **60** illustrated in FIG. **6** includes plural rotating shafts **61** to **65** attached to the frame **330** of the developing unit **33**, gears provided on the rotating shafts, a motor **66**, and a solenoid switch **67**. The rotating shaft **61** is a motor shaft of the motor **66**. A gear **611** and a gear **612** are provided on the rotating shaft **61**. The rotating shaft **62** is a drive shaft that drives the first to third transport units **334** to **336**. A gear **621** is provided on the rotating shaft **62**. The gear **621** is attached to the rotating shaft **62** by a one-way clutch. The rotating shaft **63** is a rotating shaft for transmitting the driving force. A gear **631** is provided on the rotating shaft **63**. The gear **631** is attached to the rotating shaft **63** by a one-way clutch. The rotating shaft **64** is a rotating shaft for transmitting the driving force. A gear **641** is provided on the rotating shaft **64**. The rotating shaft **64** is attached to the solenoid switch **67**. The rotating shaft **65** is a drive shaft that drives the developing roller **332**. A gear **651** and a gear **652** are provided on the rotating shaft **65**.
(50) The motor **66** generates the driving force for driving the developing roller **332** and the first to third transport units **334** to **336**. The solenoid switch **67** includes a solenoid **67***a* and is switched ON (the solenoid **67***a* is energized) or OFF (the solenoid **67***a* is deenergized) so that the rotating shaft **64** moves backward or forward in an axial direction. In the example illustrated in FIGS. **6** and **7**, the solenoid **67***a* contracts to move the rotating shaft **64** backward (FIG. **7**) when the solenoid switch **67** is ON, and expands to move the rotating shaft **64** forward (FIG. **6**) when the solenoid switch **67** is OFF.
(51) The driving force is transmitted along two paths. One path transmits the driving force from the gear **611** on the rotating shaft **61** to the gear **621** on the rotating shaft **62**, the gear **641** on the rotating shaft **64**, and the gear **651** on the rotating shaft **65** in that order. This path is referred to as a first path. The other path transmits the driving force from the gear **612** on the rotating shaft **61** to the gear **631** on the rotating shaft **63** and the gear **652** on the rotating shaft **65** in that order. This path is referred to as a second path. The first path has the one-way clutch attached to the gear **621**, and therefore the rotating shaft **62**, which is the drive shaft of the first to third transport units **334** to **336**, rotates only in one direction. When the solenoid switch **67** is ON, the rotating shaft **64** is moved backward so that the gear **641** is not in contact with the gear **621** and the gear **651**, and accordingly, the driving force is not transmitted to the rotating shaft **65**, which is the drive shaft of the developing roller **332**. The second path has the one-way clutch attached to the gear **631**, and therefore the rotating shaft **65**, which is the drive shaft of the developing roller **332**, rotates only in one direction.
(52) The drive mechanism **60** performs three types of operations: an operation in the first operation mode in which the developing roller **332** and the first to third transport units **334** to **336** are simultaneously driven; an operation in the second operation mode in which only the developing roller **332** is driven; and an operation in the second operation mode in which only the first to third transport units **334** to **336** are driven. The drive mechanism **60** illustrated in FIGS. **6** and **7** performs these operations by changing the rotation direction of the motor **66** and switching ON or OFF the solenoid switch **67**. Assume that the motor **66** rotates in a counterclockwise (CCW) direction to drive the rotating shaft **62** (drive shaft of the first to third transport units **334** to **336**)

through the one-way clutch on the gear **621**. The motor **66** rotates in a clockwise (CW) direction to rotate the gear **652** through the one-way clutch on the gear **631** and thereby drive the rotating shaft **65** (drive shaft of the developing roller **332**).

(53) In the operation in the first operation mode for image formation, the developing roller **332** and the first to third transport units **334** to **336** are driven simultaneously. In this case, the motor **66** rotates in the CCW direction, and the solenoid switch **67** is set to OFF (see FIG. **6**). On the first path, the driving force of the motor **66** is transmitted from the rotating shaft **61** to the rotating shaft **62** through the gear **611** and the gear **621** to drive the first to third transport units **334** to **336**, and is further transmitted from the gear **621** to the rotating shaft **65** through the gear **641** and the gear **651** to drive the developing roller **332**. On the second path, the transmission of the driving force is interrupted by the one-way clutch on the gear **631** when the motor **66** rotates in the CCW direction. Therefore, the second path does not contribute to the transmission of the driving force for driving the developing roller **332**.

(54) In the operation in the second operation mode for cleaning, the developing roller **332** and the first to third transport units **334** to **336** are driven individually. When only the developing roller **332** is to be driven, the motor **66** rotates in the CW direction. On the first path, when the motor **66** rotates in the CW direction, the transmission of the driving force is interrupted by the one-way clutch on the gear **621**. Therefore, the first path does not contribute to the transmission of the driving force for driving the first to third transport units **334** to **336** and the developing roller **332**. On the first path, the above-described operation is not affected by whether the solenoid switch **67** is ON or OFF. On the second path, the driving force of the motor **66** is transmitted from the rotating shaft **61** to the rotating shaft **65** through the gears **612**, **631**, and **652**, and drives the developing roller **332**. Since the driving force is transmitted to drive the first to third transport units **334** to **336** only along the first path, the first to third transport units **334** to **336** are not driven when the motor **66** rotates in the CW direction.

(55) When only the first to third transport units **334** to **336** are to be driven, the motor **66** rotates in the CCW direction, and the solenoid switch **67** is set to ON (see FIG. **7**). On the first path, the driving force of the motor **66** is transmitted from the rotating shaft **61** to the rotating shaft **62** through the gear **611** and the gear **621** to drive the first to third transport units **334** to **336**. However, since the solenoid switch **67** is ON and the gear **641** is moved backward, the driving force is not transmitted from the gear **621** to the gear **651**, and the developing roller **332** is not driven. On the second path, the transmission of the driving force is interrupted by the one-way clutch on the gear **631** when the motor **66** rotates in the CCW direction. Therefore, the second path does not contribute to the transmission of the driving force for driving the developing roller **332**.

(56) As described above, the rotation direction of the motor **66** and the ON/OFF state of the solenoid switch **67** are controlled to perform the operation in the first operation mode, the operation of driving only the developing roller **332** in the second operation mode, and the operation of driving only the first to third transport units **334** to **336** in the second operation mode. The structure of the drive mechanism **60** illustrated in FIGS. **6** and **7** is merely an example. Other structures may be applied as long as the driving force of the motor may be distributed to the developing roller **332** and the first to third transport units **334** to **336** to drive the developing roller **332** and the first to third transport units **334** to **336** simultaneously or to drive either the developing roller **332** or the first to third transport units **334** to **336**.

(57) While an exemplary embodiment of the present disclosure has been described, the technical scope of the present disclosure is not limited to the above-described exemplary embodiment. The present disclosure includes various modifications and structural replacements made without departing from the technical scope of the present disclosure.

(58) The foregoing description of the exemplary embodiments of the present disclosure has been provided for the purposes of illustration and description. It is not intended to be exhaustive or to limit the disclosure to the precise forms disclosed. Obviously, many modifications and variations

will be apparent to practitioners skilled in the art. The embodiments were chosen and described in order to best explain the principles of the disclosure and its practical applications, thereby enabling others skilled in the art to understand the disclosure for various embodiments and with the various modifications as are suited to the particular use contemplated. It is intended that the scope of the disclosure be defined by the following claims and their equivalents.

Appendix

(59) (((1)))

(60) A developing device including: a developer holder that supplies developer to an image carrier on which an electrostatic latent image is formed; a removing member that removes the developer that has not been supplied to the image carrier and remains on the developer holder; a transport unit that includes a helical rotating body, the transport unit rotating the rotating body to transport the developer and supply the developer to the developer holder; an accommodating unit that accommodates the developer holder, the removing member, and the transport unit; and a drive mechanism that operates the developer holder and the transport unit together in a first operation mode for image formation and operates the developer holder and the transport unit individually in a second operation mode for cleaning an inside of the accommodating unit by removing the developer.

(((2)))

(61) The developing device according to (((1))), wherein, in the second operation mode, the drive mechanism operates the developer holder and then operates the transport unit while a discharge port provided in the accommodating unit is opened, thereby discharging the developer through the discharge port.

(62) (((3)))

(63) The developing device according to (((2))), wherein, in the second operation mode, the drive mechanism operates the developer holder while the transport unit is stopped.

(64) (((4)))

(65) The developing device according to any one of (((1))) to (((3))), wherein, in the second operation mode, the drive mechanism operates the rotating body of the transport unit in a forward direction and a reverse direction a predetermined number of times, and then operates the transport unit while a discharge port provided in the accommodating unit is opened, thereby discharging the developer through the discharge port.

(66) (((5)))

(67) The developing device according to (((4))), wherein, in the second operation mode, the drive mechanism operates the transport unit while the developer holder is stopped at least when the rotating body of the transport unit is operated in the forward direction and the reverse direction the predetermined number of times.

(68) (((6)))

(69) The developing device according to any one of (((1))) to (((5))), further including: a developer relay unit on which the developer transported by the transport unit accumulates and that supplies the developer to the developer holder, wherein the drive mechanism operates the developer holder, the developer relay unit, and the transport unit together in the first operation mode and operates the developer holder, the developer relay unit, and the transport unit individually in the second operation mode.

(((7)))

(70) The developing device according to any one of (((1))) to (((6))), wherein, in the second operation mode, the drive mechanism operates the developer relay unit, operates the developer holder, and then operates the transport unit while a discharge port provided in the accommodating unit is opened, thereby discharging the developer through the discharge port.

(71) (((8)))

(72) An image forming apparatus including the developing device according to any one of (((1))) to (((7))).

Claims

1. A developing device comprising: a developer holder configured to supply developer to an image carrier on which an electrostatic latent image is formed; a removing member configured to remove the developer that has not been supplied to the image carrier and remains on the developer holder; a transport unit comprising a helical rotating body, wherein the transport unit is configured to rotate the helical rotating body to transport the developer and supply the developer to the developer holder; an accommodating unit that accommodates the developer holder, the removing member, and the transport unit; and a drive mechanism configured to operate the developer holder and the transport unit together in a first operation mode for image formation and to operate the developer holder and the transport unit individually in a second operation mode for cleaning an inside of the accommodating unit by removing the developer, wherein the drive mechanism is configured to, in the second operation mode, operate the helical rotating body of the transport unit in a forward direction and a reverse direction a predetermined number of times, and then operate the transport unit while a discharge port provided in the accommodating unit is opened, thereby discharging the developer through the discharge port, and wherein the drive mechanism is configured to, in the second operation mode, operate the transport unit while the developer holder is stopped at least when the helical rotating body of the transport unit is operated in the forward direction and the reverse direction the predetermined number of times.
2. The developing device according to claim 1, wherein the drive mechanism is configured to, in the second operation mode, operate the developer holder, and then operate the transport unit while a discharge port provided in the accommodating unit is opened, thereby discharging the developer through the discharge port.
3. The developing device according to claim 2, wherein the drive mechanism is configured to, in the second operation mode, operate the developer holder while the transport unit is stopped.
4. An image forming apparatus comprising: the developing device according to claim 3.
5. An image forming apparatus comprising: the developing device according to claim 2.
6. The developing device according to claim 1, further comprising: a developer relay unit configured to accumulate the developer transported by the transport unit and to supply the developer to the developer holder, wherein the drive mechanism is configured to operate the developer holder, the developer relay unit, and the transport unit together in the first operation mode and to operate the developer holder, the developer relay unit, and the transport unit individually in the second operation mode.
7. The developing device according to claim 6, wherein the drive mechanism is configured to, in the second operation mode, operate the developer relay unit, the developer holder, and then operate the transport unit while a discharge port provided in the accommodating unit is opened, thereby discharging the developer through the discharge port.
8. An image forming apparatus comprising: the developing device according to claim 7.
9. An image forming apparatus comprising: the developing device according to claim 6.
10. An image forming apparatus comprising: the developing device according to claim 1.
11. A developing device comprising: a developing roller configured to supply developer to an image carrier on which an electrostatic latent image is formed; a scraper configured to remove the developer that has not been supplied to the image carrier and remains on the developing roller; a helical rotating body, wherein the helical rotating body is configured to rotate to transport the developer and supply the developer to the developing roller; a container that accommodates the developing roller, the scraper, and the helical rotating body; and a drive mechanism configured to operate the developing roller and the helical rotating body together in a first operation mode for

image formation and to operate the developing roller and the helical rotating body individually in a second operation mode for cleaning an inside of the container by removing the developer, wherein the drive mechanism is configured to, in the second operation mode, operate the helical rotating body in a forward direction and a reverse direction a predetermined number of times, and then operate the helical rotating body while a discharge port provided in the container is opened, thereby discharging the developer through the discharge port, and wherein the drive mechanism is configured to, in the second operation mode, operate the helical rotating body while the developer holder is stopped at least when the helical rotating body is operated in the forward direction and the reverse direction the predetermined number of times.